

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Summer 2022 - 23 Examination

Semester: 8
Subject Code: 203105483
Subject Name: Data Science

Date: 22/03/2023
Time: 10:30am to 1:00pm
Total Marks: 60

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions - (All are compulsory) (Each of one mark) (15)

1. Linear Regression is a supervised machine learning algorithm. TRUE/FALSE
2. What is data science in simple words?
3. _____ measures how far each number in the set is from the mean.
4. What is CLT sample size?
5. What is bias in data science?
6. List out various types of data.
7. Define Sampling.
8. What is outlier?
9. Covariance is a statistical tool that is used to determine the relationship between the movements of two random variables. TRUE/FALSE
10. What is Information Gain?
11. Define Central tendency.
12. List graphical techniques used to visualize relationship between the variables.
13. State formula for recall.
14. Enlist any one type of data encoding techniques.
15. Enlist any two python libraries used for data visualization.

Q.2 Answer the following questions. (Attempt any three) (15)

- A) Write short note on: Encodings in Data Visualization
- B) Differentiate between Classification and Regression.
- C) Explain Data visualization using Bokeh with suitable example.
- D) What are the general techniques to handle large volumes of data?

Q.3 A) What is data collection? Explain different types of data collection methods present in Data Science. (07)

- B) What do you mean by Support Vector Machine? Explain the following terms used in mathematical representation: a) support vectors b) hyper plane c) decision boundary d) kernel trick (08)**

OR

- B) What is Naive Bayes Algorithm? Explain Naive Bayes algorithm with suitable example. (08)**

Q.4 A) Explain the phases in Data Science Process. (07)

OR

- A) Define Data Science. Discuss any application as an example. (07)**

- B) What is Data Visualization? Explain different types of data visualization techniques used in Data Science. Also discuss the importance of data visualization in data science process. (08)**